Lahore Campus

Linear Circuit Analysis (EE1001)

Final Exam

Date: June 3rd, 2024

Course Instructor(s)

Dr. Saima Zafar | Beenish Fatima (CM)
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Total Time (Hrs):	**3**
Total Marks:	**50**
Total Questions:	**5**

Roll No ________ Section ________ Student Signature ________

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1. ***Attempt all the questions.***
2. ***Attempt all parts of the same question together.***
3. ***Show all the steps with proper circuit diagrams, and answers with proper units.***

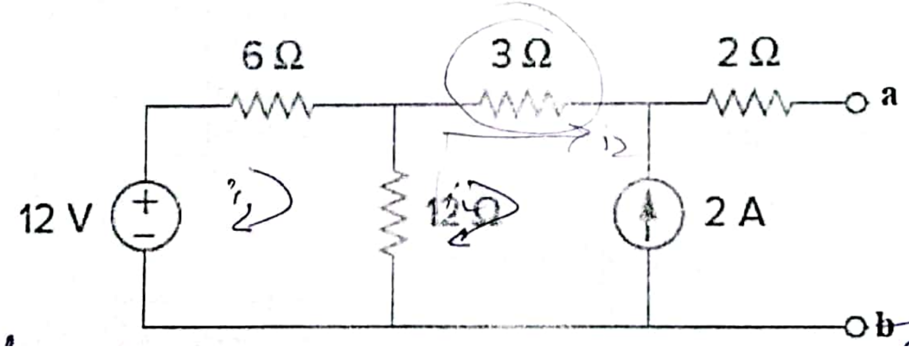
CLO # 2: Apply DC circuit analysis techniques and theorems to solve linear resistive circuits.

Q1: **[10 marks]**

a) **Solve** the circuit below to determine the Thévenin Equivalent w.r.t. terminals ‘a’ and ‘b’. **Construct** the Thévenin Equivalent circuit. **[6 marks]**

b) If a load resistor R_L is connected between terminals ‘a’ and ‘b’, **determine** the value of R_L for maximum power transfer. **[2 marks]**

c) **Calculate** the maximum power delivered to R_L. $P_{max} = \frac{V_{th}^2}{4R}$ **[2 marks]**



CLO # 3: Solve circuits containing ideal operational amplifiers

Q2: **[10 marks]**

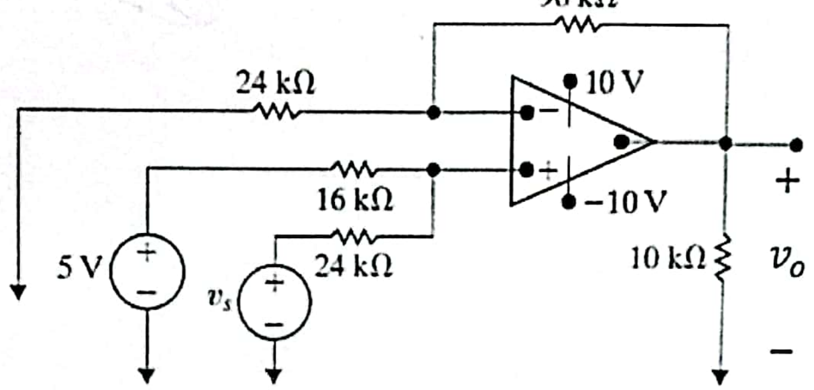
a) **Solve** the circuit to determine the expression for v_o in terms of v_s and the type of op-amp configuration. **[5 marks]**

b) **Calculate** v_o if v_s is replaced by a -3V source. **[2 marks]**

c) **Determine** the range of values for v_s such that v_o does not saturate and the op-amp remains in its linear region of operation. **[3 marks]**

$V_o = 15 + 2V_s$

$V_o = 9V \rightarrow (-3) - V_s$



CLO # 4: Analyze natural and step response of circuits containing R, L, and C

Q3: **[10 marks]**

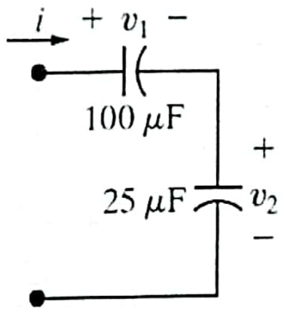
The current in the two capacitors shown below is $500\,e^{-200t}$ mA for $t \geq 0$. The initial values of v_1 and v_2 are $15\ V$ and $-10\ V$, respectively. **Figure** out the following

a) Equivalent capacitance **[2 marks]**

b) Initial energy stored in each capacitor **[2 marks]**

c) $v_1(t)$ and $v_2(t)$ for $t \geq 0$ **[4 marks]**

d) Final energy in each capacitor, as $t \to \infty$ **[2 marks]**



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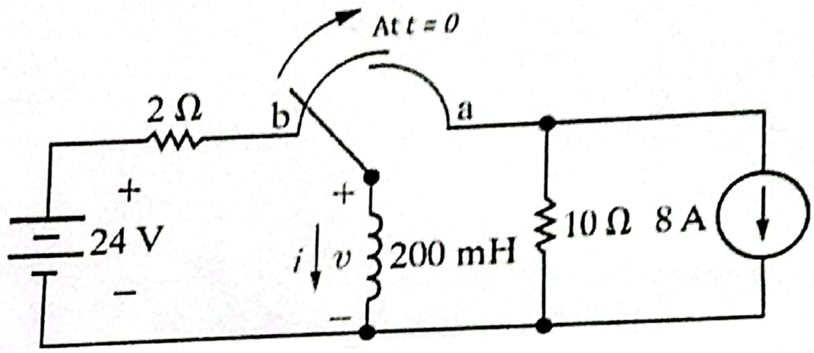
CLO # 4: Analyze natural and step response of circuits containing R, L, and C

Q4: **[10 marks]**

The switch in the circuit shown below has been in position b for a long time. At $t = 0$, the switch moves from position b to position a. **Identify** the type of response and **figure out** the following quantities:

- **a)** $i(0+)$ **[2 marks]**
- **b)** $v(0+)$ **[2 marks]**
- **c)** $\tau, t > 0$ **[2 marks]**
- **d)** $i(t), t \geq 0$ **[2 marks]**
- **e)** $v(t), t \geq 0^+$ **[2 marks]**



CLO # 2: Apply DC circuit analysis techniques and theorems to solve linear resistive circuits.

Q5: **[10 marks]**

- **a)** **Apply** the node voltage method to determine voltages at all *essential nodes*. **[7 marks]**
- **b)** **Calculate** the power associated with both dependent sources. **[3 marks]**

P=V.i